ECON573 Project First Draft

Team Bank Deposits

2023-04-03

Introduction

Our project focuses on analyzing the impact of interest rates on banks' deposit compositions and loan portfolios. In this first draft, we have compiled panel data of over one million banks' deposit compositions, which were measured quarterly from December 1992 to June 2022, comprising approximately 5,000 banks per quarter. This data was obtained from the Call Report data provided by the Federal Financial Institutions Examination Council (FFIEC). The variables included in our dataset encompass various deposit types, contemporary federal funds rates and historical four-quarter federal funds rates, bank classification types (such as commercial, savings, etc.), bank specialization groups (including consumer, mortgage, agricultural, etc.), bank regions (northeast, southwest, etc.), stock versus non-stock (raising capital through selling stock), trust power granted, and the number of insured branches. All variables have been converted to real terms adjusted for inflation.

In the current phase, we have focused on obtaining a comprehensive understanding of the deposit dataset and exploring classification techniques for analyzing the impact of interest rates on banks' deposit compositions and loan portfolios. The data organization and cleaning process for the loan portfolio will be completed later. This initial draft aims to establish a solid foundation for our analysis and to set the stage for further investigation into the complex relationships between interest rates, deposit compositions, and loan portfolios.

Data Cleaning

The code for this section is skipped

Import The Deposit Dataset

```
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(purrr)
library(plm)
```

```
##
## Attaching package: 'plm'
```

```
## The following objects are masked from 'package:dplyr':
##
##     between, lag, lead
```

```
library (MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```
library(e1071)

deposit_data = read_csv("df_panel_lag_final.csv", show_col_types = F)

# Calculate proportions
deposit_data[, 27:40] = deposit_data[, 27:40]/as.numeric(deposit_data$Total_Deposits)
```

Basic Summary Statistics

Group the dataset by bank classification types and calculate the mean and standard deviation of total deposits and different types of deposits under different classification type.

We have the following bank classification types: 1. N = commercial bank, national (federal) charter and fed member, supervised by the office of the comptroller of the currency (occ) 2. SM = commercial or savings bank, state charter and fed member, supervised by the federal reserve (FRB) 3. NM = commercial bank, state charter and fed nonmember, supervised by the FDIC or OCC 4. SB = savings banks, state charter, supervised by the FDIC 5. SA = As of July 21, 2011, FDIC supervised state chartered thrifts and OCC supervised federally chartered thrifts. Prior to that date, state or federally chartered savings associations supervised by the Office of Thrift Supervision (OTS). 6. OI = Insured U.S. branch of a foreign chartered institution (IBA)

```
bkclass_summary <- deposit_data %>%
  group_by(bkclass) %>%
  summarise(
    count = n(),
    across(contains("Deposits"), list(mean = ~mean(., na.rm = TRUE), sd = ~sd(., na.rm = TRUE)), .names
  )

head(bkclass_summary)
```

```
## # A tibble: 6 x 32
##   bkclass  count Total_Deposits_mean Total_Deposits_sd Deposits_Domestic_mean
##   <chr>    <int>               <dbl>             <dbl>                  <dbl>
## 1 N       209315              12946.           187246.                  0.997
## 2 NM      543904               1448.            11523.                  0.999
## 3 OI        1809               6450.            18186.                  0.927
## 4 SA      101500               3678.            24002.                  1.00
## 5 SB       52249               2624.             7702.                  1.00
## 6 SM      105967               7145.            52292.                  0.995
## # ... with 27 more variables: Deposits_Domestic_sd <dbl>,
## #   Deposits_Foreign_mean <dbl>, Deposits_Foreign_sd <dbl>,
## #   Demand_Deposits_mean <dbl>, Demand_Deposits_sd <dbl>,
## #   Money_Market_Deposits_mean <dbl>, Money_Market_Deposits_sd <dbl>,
## #   Saving_Deposits_mean <dbl>, Saving_Deposits_sd <dbl>,
## #   Time_Deposits_mean <dbl>, Time_Deposits_sd <dbl>,
## #   Total_Time_and_Saving_Deposits_mean <dbl>, ...
```

Group the dataset by bank specialization groups and calculate the mean and standard deviation of total deposits and different types of deposits under different groups.

We have the following groups: 1. International Specialization Institutions with assets greater than $10 billion and more than 25 percent of total assets in foreign offices. 2. Agricultural Specialization Banks with agricultural production loans plus real estate loans secured by farmland in excess of 25 percent of total loans and leases. 3. Credit-card Specialization Institutions with credit-card loans plus securitized receivables in excess of 50 percent of total assets plus securitized receivables. 4. Commercial Lending Specialization Institutions with commercial and industrial loans, plus real estate construction and development loans, plus loans secured by commercial real estate properties in excess of 25 percent of total assets. 5. Mortgage Lending Specialization Institutions with residential mortgage loans, plus mortgage-backed securities, in excess of 50 percent of total assets. 6. Consumer Lending Specialization Institutions with residential mortgage loans, plus credit-card loans, plus other loans to Individuals, in excess of 50 percent of total assets. 7. Other Specialized $1 Billion Institutions with assets greater than $1 billion that do not meet any of the definitions above, they have significant lending activity with no identified asset concentrations.

```r
specgrp_summary <- deposit_data %>%
  group_by(specgrp) %>%
  summarise(
    count = n(),
    across(contains("Deposits"), list(mean = ~mean(., na.rm = TRUE), sd = ~sd(., na.rm = TRUE)), .names
  )
head(specgrp_summary)
```

```
## # A tibble: 6 x 32
##   specgrp  count Total_Deposits_mean Total_Deposits_sd Deposits_Domestic_mean
##     <dbl>  <int>               <dbl>             <dbl>                  <dbl>
## 1       1    775            1268829.          1779713.                  0.480
## 2       2 216014                402.              854.                  1
## 3       3   4241              24287.            62761.                  0.977
## 4       4 434129                4163.            31568.                  0.998
## 5       5 117606                3183.            21573.                  1.00
## 6       6  18972                3288.            22984.                  0.999
## # ... with 27 more variables: Deposits_Domestic_sd <dbl>,
## #   Deposits_Foreign_mean <dbl>, Deposits_Foreign_sd <dbl>,
## #   Demand_Deposits_mean <dbl>, Demand_Deposits_sd <dbl>,
## #   Money_Market_Deposits_mean <dbl>, Money_Market_Deposits_sd <dbl>,
```

```
## #   Saving_Deposits_mean <dbl>, Saving_Deposits_sd <dbl>,
## #   Time_Deposits_mean <dbl>, Time_Deposits_sd <dbl>,
## #   Total_Time_and_Saving_Deposits_mean <dbl>, ...
```

Group the dataset by stock versus non-stock (raising capital through selling stock) and calculate the mean and standard deviation of total deposits and different types of deposits under different groups.

1 = Stock 0 = Non-stock

```
mutual_summary <- deposit_data %>%
  group_by(mutual) %>%
  summarise(
    count = n(),
    across(contains("Deposits"), list(mean = ~mean(., na.rm = TRUE), sd = ~sd(., na.rm = TRUE)), .names
  )

head(mutual_summary)
```

```
## # A tibble: 2 x 32
##   mutual     count Total_Deposits_mean Total_Deposits_sd Deposits_Domestic_mean
##   <chr>      <int>               <dbl>             <dbl>                  <dbl>
## 1 Non-Stock  72436               1052.             1977.                  1
## 2 Stock     942308               4988.            90879.                  0.998
## # ... with 27 more variables: Deposits_Domestic_sd <dbl>,
## #   Deposits_Foreign_mean <dbl>, Deposits_Foreign_sd <dbl>,
## #   Demand_Deposits_mean <dbl>, Demand_Deposits_sd <dbl>,
## #   Money_Market_Deposits_mean <dbl>, Money_Market_Deposits_sd <dbl>,
## #   Saving_Deposits_mean <dbl>, Saving_Deposits_sd <dbl>,
## #   Time_Deposits_mean <dbl>, Time_Deposits_sd <dbl>,
## #   Total_Time_and_Saving_Deposits_mean <dbl>, ...
```

Group the dataset by banks' location (state) and calculate the mean and standard deviation of total deposits and different types of deposits under different groups.

```
stalp_summary <- deposit_data %>%
  group_by(stalp) %>%
  summarise(
    count = n(),
    across(contains("Deposits"), list(mean = ~mean(., na.rm = TRUE), sd = ~sd(., na.rm = TRUE)), .names
  )

# Display the summary statistics
head(stalp_summary)
```

```
## # A tibble: 6 x 32
##   stalp count Total_Deposits_mean Total_Deposits_sd Deposits_Domestic_mean
##   <chr> <int>               <dbl>             <dbl>                  <dbl>
## 1 AK      826               3390.             5234.                  1.00
## 2 AL    18701               4525.            31854.                  0.999
## 3 AR    19243               1372.             5145.                  1.00
## 4 AS       73                361.               87.5                1
## 5 AZ     4282               3156.            12363.                  1.00
## 6 CA    35801               8258.            44188.                  0.996
```

```
## # ... with 27 more variables: Deposits_Domestic_sd <dbl>,
## #   Deposits_Foreign_mean <dbl>, Deposits_Foreign_sd <dbl>,
## #   Demand_Deposits_mean <dbl>, Demand_Deposits_sd <dbl>,
## #   Money_Market_Deposits_mean <dbl>, Money_Market_Deposits_sd <dbl>,
## #   Saving_Deposits_mean <dbl>, Saving_Deposits_sd <dbl>,
## #   Time_Deposits_mean <dbl>, Time_Deposits_sd <dbl>,
## #   Total_Time_and_Saving_Deposits_mean <dbl>, ...
```

Generate Categorical Variables

The dataset contains the proportion of each deposit type, which are not categorical variables. However, classification techniques require that the response variable has to be categorical. Therefore, we create several categorical variables that we are interested in:

1. Is the proportion of demand deposits (proxy for volatile funding source) higher than saving deposits (proxy for predictable funding source)?

```
demand_vs_saving = as.factor((deposit_data$Demand_Deposits>=deposit_data$Saving_Deposits)*1)
```

2. Is the proportion of time deposits (proxy for stable and expensive funding source) higher than non-time deposits (proxy for volatile and cheap funding source)?

```
time_vs_nontime = as.factor((deposit_data$Time_Deposits>=0.5)*1)
```

3. Is the proportion of interest-bearing deposits (proxy for expensive funding source) higher than non-interest-bearing deposits (proxy for cheap funding source)?

```
intb_vs_nonintb = as.factor((deposit_data$Interest_Bearing_Deposits>=deposit_data$`Non-interest_Bearing
```

4. Is the proportion of retail deposits (usually in small denomination) higher than brokered deposits (usually in large denomination)?

```
retail_vs_brokered = as.factor((deposit_data$Retail_Deposits>=deposit_data$Brokered_Deposits)*1)
```

5. Is the proportion of insured deposits (secured creditors) higher than non-insured deposits (unsecured creditors)?

```
insured_vs_notinsured = as.factor((deposit_data$Estimated_Insured_Deposits>=0.5)*1)
```

Then we combine the original dataset with these new categorical variables.

```
deposit_data_new = cbind(deposit_data, demand_vs_saving, time_vs_nontime, intb_vs_nonintb, retail_vs_br

deposit_data_new_pdata = pdata.frame(deposit_data_new, index = c("cert"))
rownames(deposit_data_new_pdata) = 1:nrow(deposit_data_new_pdata)
```

Validation Set

```r
set.seed(1)
train <- sample(c(TRUE, FALSE), nrow(deposit_data_new_pdata), rep = TRUE)
test <- (!train)

training_sample <- deposit_data_new_pdata[train,]
testing_sample <- deposit_data_new_pdata[test,]
```

Classification

In this section we take "time deposits vs non-time deposit" as an example to examine the performance of logistic regression, QDA, and Naive Bayes.

Logistic Regression

```r
logit_time_vs_nontime = glm(time_vs_nontime~ffr_0+ffr_1+ffr_2+ffr_3+ffr_4
                            +as.factor(time),
                            family = "binomial",
                            data = training_sample[!is.na(training_sample$time_vs_nontime) &
                                        !is.nan(training_sample$time_vs_nontime) &
                                        !is.infinite(training_sample$time_vs_nontime), ])

# Make predictions on the testing sample
predicted_probabilities <- predict(logit_time_vs_nontime, newdata = testing_sample, type = "response")

# Convert probabilities to binary predictions using a threshold of 0.5
predicted_classes <- ifelse(predicted_probabilities > 0.5, 1, 0)

# Create a confusion matrix
confusion_matrix <- table(Predicted = predicted_classes, Actual = testing_sample$time_vs_nontime)

# Display the confusion matrix
print(confusion_matrix)
```

```
##          Actual
## Predicted      0      1
##         0 243952 116816
##         1  54535  61922
```

```r
# Calculate the error rate
total_cases <- sum(confusion_matrix)  # Total number of cases
correct_cases <- sum(diag(confusion_matrix))  # Correctly classified cases
error_rate <- 1 - (correct_cases / total_cases)
cat("Error rate:", error_rate, "\n")
```

```
## Error rate: 0.359057
```

QDA

```
qda_time_vs_nontime = qda(time_vs_nontime~ffr_0+ffr_1+ffr_2+ffr_3+ffr_4
                          +as.factor(time),
                          data = training_sample[!is.na(training_sample$time_vs_nontime) &
                                       !is.nan(training_sample$time_vs_nontime) &
                                       !is.infinite(training_sample$time_vs_nontime), ])

# Predictions
qda.pred <- predict(qda_time_vs_nontime, testing_sample)

# Confusion Matrix
qda.class <- qda.pred$class
confusion_matrix = table(qda.class, testing_sample$time_vs_nontime)

# Display the confusion matrix
print(confusion_matrix)
```

```
##
## qda.class      0      1
##         0 124899  40141
##         1 173588 138597
```

```
# Calculate the error rate
total_cases <- sum(confusion_matrix)  # Total number of cases
correct_cases <- sum(diag(confusion_matrix))  # Correctly classified cases
error_rate <- 1 - (correct_cases / total_cases)
cat("Error rate:", error_rate, "\n")
```

```
## Error rate: 0.4478579
```

Naive Bayes

```
nb_time_vs_nontime = naiveBayes(time_vs_nontime~ffr_0+ffr_1+ffr_2+ffr_3+ffr_4
                          +as.factor(time),
                          data = training_sample[!is.na(training_sample$time_vs_nontime) &
                                       !is.nan(training_sample$time_vs_nontime) &
                                       !is.infinite(training_sample$time_vs_nontime), ])

# Predictions
nb.class <- predict(nb_time_vs_nontime, testing_sample)

# Confusion Matrix
confusion_matrix = table(nb.class, testing_sample$time_vs_nontime)

# Display the confusion matrix
print(confusion_matrix)
```

```
##
## nb.class      0      1
##        0 193952  74495
##        1 104535 104243
```

```r
# Calculate the error rate
total_cases <- sum(confusion_matrix)  # Total number of cases
correct_cases <- sum(diag(confusion_matrix))  # Correctly classified cases
error_rate <- 1 - (correct_cases / total_cases)
cat("Error rate:", error_rate, "\n")
```

```
## Error rate: 0.375148
```